

SRI ESHWAR COLLEGE OF ENGINEERING

Department of Computer Science and Engineering

INTELLIDATA 2026

DATA SCIENCE HACKATHON - DAY 2

STOCKSENSE

Predict Demand. Prevent Stock-outs. Power Better Decisions.

INDUSTRY USE CASE: NOVAMART RETAIL PVT. LTD.

Your team is the Data Science consulting unit for a multi-city retail company. Management wants a working decision-support solution - not just a model - that can forecast demand, predict stock-out risk, explain the reasons, and recommend an inventory action.

Event	Details
Event	IntelliData 2026 - Data Science Hackathon
Institution	Sri Eshwar College of Engineering, Coimbatore
Department	Computer Science and Engineering
Participants	III Year CSE
Team Size	Maximum 3 students per team
Submission	GitHub repository

The Business Question

Can we predict what will sell, identify products likely to go out of stock, and tell store managers what action to take before the problem occurs?

Mandatory Hackathon Rounds

Round	Mandatory Task
Round 1	Data Understanding, Preparation & EDA
Round 2	Feature Engineering & Machine Learning
Round 3	Intelligence Layer, Visualization & Prototype

Hackathon Mission

- Forecast the next 7-day demand for each Store x Product combination.
- Predict the probability of stock-out and classify risk as High / Medium / Low.
- Create a replenishment recommendation with a clear business explanation.
- Build a management-ready visualization/prototype and present the final solution.

Challenge Line: “Every empty shelf is lost revenue. Can Data Science predict it before it happens?”

1. BUSINESS CONTEXT & SUCCESS CRITERIA

NovaMart Retail Pvt. Ltd. operates supermarkets across multiple cities and sells groceries, beverages, dairy, snacks, personal care, household and frozen products. Replenishment decisions are currently based on simple rules and manager experience. The company faces two costly problems: stock-outs that cause lost sales and overstocking that locks working capital and may lead to expiry.

Business Objectives

- Reduce stock-out incidents and lost sales.
- Reduce excess inventory and holding cost.
- Improve product availability and customer satisfaction.
- Create a repeatable data-driven replenishment process.
- Give managers a simple, explainable action list.

Key Business KPIs

KPI	Meaning / Suggested Calculation
Revenue	Sum (Quantity x Selling Price)
Units Sold	Sum Quantity
Stock-out Rate	Stock-out events / total product-store observations
Inventory Turnover	Cost of goods sold / average inventory value
Days of Inventory	Closing stock / average daily demand
Promotion Lift	(Promotional sales - normal sales) / normal sales
Estimated Lost Sales	Unfulfilled demand x selling price

Solution Architecture

Raw Data -> Clean & Merge -> EDA -> Statistics -> Features -> ML Models -> Explainability -> Recommendation -> Visualization / Prototype

Mandatory Modelling Problems

Model	Target / Purpose
Model 1 - Demand Forecasting	Target: next_7_day_demand. Regression problem to estimate demand before the next replenishment cycle.
Model 2 - Stock-out Risk	Target: stockout_flag / stockout_probability. Classification problem to identify high-risk Store x Product combinations.

Business Action Layer

The final system must turn model outputs into decisions. A prediction without an action is not considered complete. The solution should produce a recommended reorder quantity, risk level, key reasons and a manager-friendly recommendation.

2. DATASETS - FORMAT & SAMPLE DATA (PART 1)

Teams will work with multiple CSV files rather than one perfectly prepared table. This simulates how company data is stored across sales, product, store, inventory and external systems.

A. transactions.csv

Field	Type	Description
transaction_id	string	Unique transaction reference
date	date	Transaction date
store_id	string	Store identifier
product_id	string	Product identifier
quantity	integer	Units sold
selling_price	float	Actual unit selling price
discount_pct	float	Discount percentage
promotion_flag	0/1	Whether a promotion was active
customer_id	string	Customer identifier
payment_mode	category	UPI / Card / Cash
hour	integer	Hour of transaction (0-23)

Sample Rows

transaction_id	date	store_id	product_id	qty	price	disc%	promo	payment	hour
T1001	2026-08-01	S01	P101	2	58	0	0	UPI	09
T1002	2026-08-01	S01	P205	1	110	10	1	Card	10
T1003	2026-08-01	S02	P101	3	58	0	0	Cash	11
T1004	2026-08-02	S01	P330	4	42	15	1	UPI	18

B. products.csv

Field	Description
product_id	Product identifier
category	Product category
sub_category	Sub-category
brand	Brand
mrp	Maximum retail price
cost_price	Procurement cost
shelf_life_days	Expected shelf life
supplier_id	Primary supplier

product_id	category	sub_category	brand	mrp	cost	shelf_life	supplier_id
P101	Dairy	Milk	Aavin	60	52	3	SUP01
P205	Personal Care	Shampoo	GlowCare	120	79	730	SUP08
P330	Beverages	Soft Drink	FizzUp	50	31	180	SUP04
P442	Snacks	Biscuits	Crispo	35	22	240	SUP06

3. DATASETS - FORMAT & SAMPLE DATA (PART 2)

C. stores.csv

store_id	city	store_type	floor_area_sqft	avg_daily_customers	region
S01	Coimbatore	Supermarket	8500	1250	West
S02	Chennai	Hypermarket	18000	2850	North
S03	Madurai	Express	4200	720	South
S04	Salem	Supermarket	7600	1080	Central

D. inventory.csv

date	store	product	opening	received	sold	closing	reorder_lvl	lead_days
2026-08-01	S01	P101	310	100	95	315	220	1
2026-08-01	S01	P205	125	0	18	107	60	4
2026-08-01	S02	P101	520	150	160	510	350	1
2026-08-01	S03	P330	210	80	120	170	140	3
2026-08-02	S01	P330	260	0	145	115	160	3

E. external_factors.csv

date	city	temp_c	rain_mm	holiday	festival	weekend	local_event
2026-08-01	Coimbatore	29.1	0	0	0	1	0
2026-08-01	Chennai	33.4	2.5	0	0	1	1
2026-08-02	Coimbatore	28.7	6.2	0	0	1	0
2026-08-15	Coimbatore	27.9	0	1	0	1	1
2026-08-25	Madurai	34.0	0	0	1	0	0

Intentional Data Quality Traps

Issue	Example	What the team should do
Missing values	temperature = blank	Impute or justify exclusion
Duplicate rows	same transaction_id repeated	Identify and remove true duplicates
Category inconsistency	Beverages / beverage / BEVERAGES	Standardise categories
Impossible quantity	quantity = -4	Validate, flag and treat appropriately
Inventory mismatch	closing != opening + received - sold	Detect and reconcile / flag
Sparse history	new product with only 5 days data	Use fallback strategy / explain limitation

Expected Master Table Grain

ONE ROW = ONE DATE x ONE STORE x ONE PRODUCT

After aggregation and merging, create a master analytics dataset containing demand, stock, price, promotion, store, product and external variables at the same grain.

4. ROUND 1 - DATA UNDERSTANDING, PREPARATION & EDA

Data Cleaning & Integration

- Audit data types, missing values, duplicates and invalid values.
- Standardise product categories, dates and IDs.
- Validate inventory arithmetic wherever possible.
- Aggregate transactions to daily Store x Product level.
- Merge product, store, inventory and external-factor tables.
- Prepare a short Data Quality Report: issue, count, action taken and justification.

Exploratory Data Analysis (EDA)

Produce meaningful business insights. Every chart must answer a business question.

Business Question	Suggested Analysis
Which categories generate most revenue?	Category share + Pareto analysis
Which stores are growing or declining?	Daily/weekly trend by store
Do promotions increase units sold?	Promotion vs non-promotion comparison
How does weekend behaviour differ?	Weekday vs weekend demand
Which products are volatile?	Coefficient of variation / rolling standard deviation
Which stores repeatedly stock out?	Stock-out heatmap by store and category

Statistical Reasoning

Perform at least 3 statistical analyses. State the business question, H0, H1, test used, p-value and business interpretation.

- Do promotions significantly increase sales?
- Does mean demand differ across store types?
- Is stock-out frequency associated with promotion status?

5. ROUND 2 - FEATURE ENGINEERING & MACHINE LEARNING

Feature Engineering

Feature Group	Examples
Time	day_of_week, weekend_flag, month, week_no, festival_flag
Lag	lag_1, lag_7, lag_14 demand
Rolling	rolling_mean_7, rolling_mean_14, rolling_std_7
Inventory	days_of_inventory, inventory_to_demand_ratio, reorder_gap
Price / Promo	discount_pct, price_change, promotion_flag
Store / Product	store_type, category, brand, shelf_life, lead_time

Rule: Avoid data leakage. Any feature used to predict the future must contain only information available at prediction time.

Permitted ML Algorithms for IntelliData 2026

Use Machine Learning only. Deep Learning models are not permitted. Select appropriate algorithms from the following fixed list:

Permitted Algorithms
Linear Regression
Decision Trees
Support Vector Machine (SVM)
XGBoost
K-Nearest Neighbours (KNN)
Naive Bayes
Random Forest

6. MACHINE LEARNING, VALIDATION & EXPLAINABILITY

Model 1 - 7-Day Demand Forecast

- Target: next_7_day_demand.
- Build a simple baseline first, such as the previous 7-day mean.
- Train and compare suitable regression algorithms from the permitted IntelliData 2026 ML list.
- Use time-aware train/validation/test splitting. Do not randomly mix future dates into training.
- Compare MAE, RMSE, MAPE and R²; explain which metric matters most to the business.

Model 2 - Stock-out Risk Classification

- Target: stockout_flag and stockout_probability.
- Create a clear operational definition of stock-out.
- Train and compare suitable classification algorithms from the permitted IntelliData 2026 ML list.
- Evaluate Accuracy, Precision, Recall, F1-score, ROC-AUC and Confusion Matrix.
- Handle class imbalance appropriately.

Suggested Business Logic

Item	Formula / Logic
Recommended Stock	Forecast Demand + Safety Stock
Reorder Quantity	$\max(0, \text{Recommended Stock} - \text{Current Stock} - \text{Incoming Stock})$
High Risk	Stock-out probability ≥ 0.70
Medium Risk	$0.40 \leq \text{probability} < 0.70$
Low Risk	Probability < 0.40

Explainability - Mandatory

Management will not accept a black-box alert. Use feature importance or permutation importance to explain why a product is at risk. The explanation should be understandable to a store manager.

Example explanation for Store S01 - Cola 750 ml	Impact
Promotion active	+31%
Weekend approaching	+22%
Recent sales growth	+19%
Festival effect	+17%
Temperature increase	+11%

7. ROUND 3 - INTELLIGENCE LAYER, VISUALIZATION & PROTOTYPE

- Convert model predictions into High / Medium / Low risk levels.
- Generate a recommended reorder quantity and manager-friendly action.
- Present key drivers behind each high-risk prediction.
- Create an interactive visualization/dashboard or simple prototype using an appropriate tool.
- Optional: build a What-If analysis for discount, supplier delay or festival-demand changes.

Expected Manager-Ready Output

Store	Product	Current Stock	7-Day Forecast	Stock-out Prob.	Risk	Recommended Order
S01	Milk 1L	250	485	0.89	HIGH	180
S01	Bread 400g	400	355	0.18	LOW	0
S03	Cola 750ml	180	390	0.82	HIGH	245
S04	Shampoo 180ml	95	108	0.46	MEDIUM	40

8. EXPECTED OUTPUTS & SUBMISSION

Example Recommendation Card

STORE S01 - MILK 1L

Predicted 7-day demand: 485 units

Current stock: 250 units | Incoming stock: 80 units

Stock-out probability: 89% | Risk: HIGH

Recommended additional replenishment: 180 units

Why? Weekend approaching, promotion running, 16% recent sales growth, local festival.

MANAGER ACTION: Raise replenishment order today.

Visualization / Prototype - Minimum Sections

Section	Must Show
Executive Summary	Revenue, growth, stock-out rate, inventory value, products at risk
Demand Intelligence	Actual vs forecast, category trend, store trend, forecast error
Inventory Risk	High / Medium / Low risk table or heatmap
Manager Action Centre	Recommended reorder quantity + reason
Model Performance	Metrics, confusion matrix, residual/error analysis
Explainability	Top features / importance interpretation

Expected Files / Deliverables

- Cleaned master dataset (CSV).
- EDA + modelling notebook(s).
- Data Quality Report.
- Model comparison table and selected-model justification.
- Saved model / reproducible pipeline.
- Visualization / dashboard / prototype.
- README with setup and usage instructions.
- Final presentation / pitch.
- GitHub repository link as instructed.

9. RECOMMENDED TOOLS

Purpose	Suggested Tools
Data Preparation	Python, Pandas, NumPy, Jupyter Notebook / Google Colab
Visualization	Matplotlib, Seaborn, Plotly; Power BI or Tableau
Statistics	SciPy, Statsmodels
Machine Learning	scikit-learn, XGBoost
Explainability	Feature Importance, Permutation Importance
Prototype	Streamlit or Plotly Dash (optional)
Version Control	Git + GitHub

Documentation	README, Markdown, PPT/PDF
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10. TEAM ROLES & EVALUATION

Student	Suggested Primary Ownership
Student 1 - Data Analyst	Cleaning, integration, EDA, statistics, data quality
Student 2 - ML Engineer	Feature engineering, forecasting, validation
Student 3 - Decision Intelligence	Classification, explainability, visualization, recommendation

All team members must understand the complete solution and may be questioned on any module during evaluation.

Evaluation Framework

Area	Weight
Business Understanding	10%
Data Cleaning & Data Quality	15%
EDA & Statistical Reasoning	15%
Feature Engineering	15%
ML Modelling & Validation	20%
Explainability & Recommendation Logic	10%
Visualization / Prototype	10%
Final Pitch & Team Understanding	5%

Suggested Submission Folder Structure

STOCKSENSE_TEAM_NAME/

data/

raw/

processed/

notebooks/

src/

models/

dashboard/

reports/

README.md

requirements.txt

Final Pitch

Explain what you discovered, what you can predict, model performance, why the result can be trusted, which products/stores need action now, and how the recommendation reduces business loss. Build a business solution - not just a model.